nature materials

LETTERS

PUBLISHED ONLINE: 11 OCTOBER 2009 | DOI: 10.1038/NMAT2546

Plasmonic nanorod metamaterials for biosensing

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Label-free plasmonic biosensors rely either on surface plasmon polaritons or on localized surface plasmons on continuous or nanostructured noble-metal surfaces to detect molecular-binding events[1–4]. Despite undisputed advantages, including spectral tunability[3], strong enhancement of the local electric field[5,6] and much better adaptability to modern nanobiotechnology architectures[7], localized plasmons demonstrate orders of magnitude lower sensitivity compared with their guided counterparts[3]. Here, we demonstrate an improvement in biosensing technology using a plasmonic metamaterial that is capable of supporting a guided mode in a porous nanorod layer. Benefiting from a substantial overlap between the probing field and the active biological substance incorporated between the nanorods and a strong plasmon-mediated energy confinement inside the layer, this metamaterial provides an enhanced sensitivity to refractive-index variations of the medium between the rods (more than 30,000 nm per refractive-index unit). We demonstrate the feasibility of our approach using a standard streptavidin–biotin affinity model and record considerable improvement in the detection limit of small analytes compared with conventional label-free plasmonic devices.

Confined to a metal/dielectric interface, surface plasmons are known to be extremely sensitive to the refractive index of the dielectric medium within the penetration depth of the evanescent field[8]. This remarkable property has been used for the development of label-free plasmonic biosensors[1], which emerged as a leading modern technology for detection and studies of binding events between the target analyte and its corresponding receptor on a metal surface[7]. Most of the surface plasmon resonance (SPR) biosensors commercially available at present use surface plasmon polaritons (SPPs), which are excited on continuous Au or Ag films in the attenuated total internal reflection (ATR) geometry[2]. Owing to the resonant photon–SPP coupling conditions, namely the specific combination of the wavevector and wavelength, SPPs provide an extremely small detection limit exceeding 10^{-5} refractive-index units (RIU), which can be further improved using phase-sensitive interferometry schemes[9,10]. However, the SPP-based approach still needs an improvement in sensitivity for the detection of small analytes (normally, less than 500 Da) and does not always satisfy modern requirements of biotechnology advancing towards new nanoscale designs and promising the manipulation on the nanoscale level, size-based selectivity and selective chemical and biochemical nano-architectures[7]. Localized surface plasmons (LSPs) of metallic nanostructures seem much more suitable to match these new trends, as well as to bring new functionalities, such as spectral tunability[3,11] and strong enhancement of the local electric field[5,6]. However, LSP-based sensors are known to provide at least an order of magnitude lower sensing response to refractive-index change compared with SPPs, with sensitivities not exceeding 100–300 nm per RIU in spectral interrogation schemes as well as 10 times smaller probe depth[3,12], making them applicable for only a very limited number of biological species.

Understanding these problems in sensing applications of existing plasmonics structures, we focused our attention on newly emerging plasmonic metamaterials, composite structures consisting of subwavelength-size components, and designed a sensor-oriented metamaterial that is capable of supporting similar or more sensitive guiding modes than SPPs. The metamaterial-based approach may also offer the advantages of nano-architectures, large effective area with the possibility of guiding and imaging capabilities beyond the diffraction limit[13–15], nanomanipulation and nanotrapping[16,17]. Here, we report on the experimental realization of such a metamaterial-based transducer using an array of parallel gold nanorods oriented normally to a glass substrate. When the distance between the nanorods is smaller than the wavelength, the layer of such metamaterial supports a guided mode with the field distribution inside the layer determined by plasmon-mediated interaction between the nanorods. This anisotropic guided mode has resonant excitation conditions similar to the SPP mode of a smooth metal film, providing very high sensitivity to refractive-index change of more than 30,000 nm per RIU and a large probe depth (500 nm). The designed metamaterial outperforms localized and SPP-based sensors available at present and still keeps the advantages of nanoplasmonic structure, such as tunable spectral response, nanoscale bioimmobilization template and strong field localization inside the template matrix.

The metamaterial consists of an assembly of Au nanorods electrochemically grown into a substrate-supported, thin-film porous aluminium oxide template as described in the Supplementary Information. The final structure represents an array of parallel nanorods occupying an area of up to 2 cm^2 (Fig. 1a). The structural parameters can be controlled by altering the fabrication conditions with typical dimensions being in the range of rod lengths between 20 and 700 nm, rod diameter 10–50 nm and separation from 40 to 70 nm, thus achieving a nanorod areal density of approximately 10^{10}–10^{11} cm^{-2}. The lateral size and separations between the nanorods are much smaller than the wavelength of light used in the experiments, so only average values of nanorod assembly parameters are important, and individual nanorod size deviations have no influence on the optical properties that are well described by an effective medium model (see Supplementary Information

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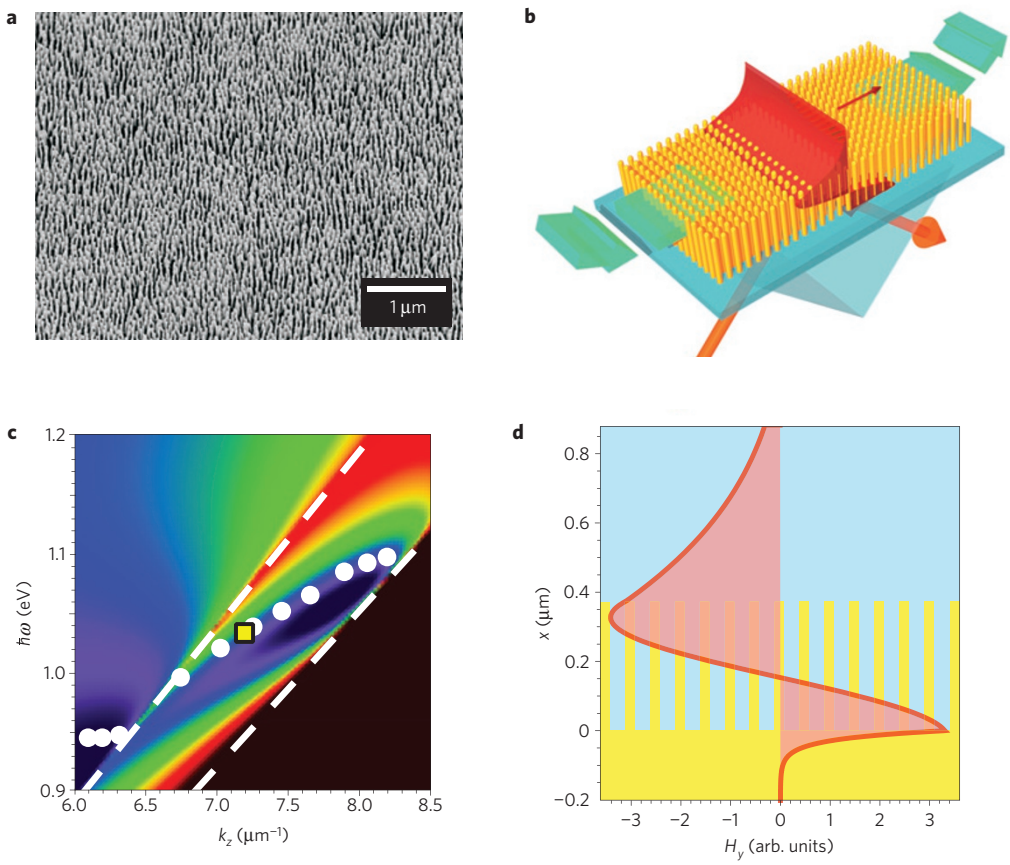

Figure 1 | Plasmonic nanorod metamaterial and the properties of the electromagnetic mode used for sensing experiments. a, Typical scanning electron micrograph of the nanorod assembly. **b**, Schematic of the ATR measurements and flow cell. **c**, Dispersion of the guided mode in the metamaterial/water system. Colour plot: numerical simulations; circles: experimental dispersion plotted from Fig. 2b; dashed lines: light lines in air (top left) and the substrate (bottom right). **d**, The calculated electromagnetic-field distribution of the guided mode in the metamaterial layer. The profile corresponds to the position indicated by the yellow square in **c**.

for details). For this reason, the optical properties of the nanorod arrays are very stable with respect to fabrication tolerances[18–21]. The fabrication technology made it possible to fully or partially embed the nanorods into an alumina matrix. The nanorod structures' sensing properties were characterized in both direct-transmission and ATR geometries using spectrometric and ellipsometric platforms (details in the Supplementary Information).

In the direct-transmission geometry, the extinction spectra of the nanorod array in an air environment show two pronounced peaks at 520 and 720 nm, as shown in Fig. 2a. These resonances correspond to the transverse and longitudinal modes of plasmonic excitations in the assembly[18–20]. The transverse mode is related to the electron motion perpendicular to the nanorod long axes and can be excited with the light having an electric-field component perpendicular to it, whereas the longitudinal mode is related to the electron oscillations along the nanorod axes and requires p-polarized light to be excited with a component of the incident electric field along the nanorods. The electric-field distribution associated with the longitudinal mode has an intensity maximum in the middle of the nanorods and results from a strong dipole–dipole interaction between the plasmons of individual nanorods in the array and is, therefore, not sensitive to the presence of the superstrate[18–21]. Under oblique incidence and with p-polarized light, the electric field couples to both the transverse and longitudinal modes with the metamaterial slab showing extreme anisotropy of dielectric permittivity similar to that of a uniaxial crystal[22]. The illumination of the same nanorod structure in the ATR geometry reveals the new guided mode in the near-infrared spectral range (Fig. 2b,c), which is excited only with p-polarized light and dominates the optical response of the assembly. The position of the dip in the reflection spectrum associated with this guided mode depends on the nanorod length, as well as on the refractive index of the environment around the rods. It is also important that conditions of excitation of this mode are strongly dependent on the angle of incidence θ and incident light polarization. Taking into account these conditions (resonant combination of θ and λ), we can plot the dispersion of this mode (Fig. 1c). The numerical analysis (details in the Supplementary Information) identifies its nature as a guided mode in an anisotropic planar waveguide formed by the layer of nanorods with a permittivity determined by the coupling between the plasmonic modes of the individual nanorods. This guided mode has unique features being a bulk mode but existing only in the thin subwavelength slabs of anisotropic metamaterial. The nature of the guided mode is confirmed by a coincidence of its calculated dispersion to experimentally measured dependence (Fig. 1c). The simulations show the field of this mode largely concentrated within the metamaterial but also substantially extended in the superstrate (Fig. 1d). To further investigate the nature of this mode, we examined a gold-nanorod-based structure, which was completely embedded into an alumina matrix. The alumina matrix was intended to protect all modes with the electric field localized inside the nanorod layer from the impact of the adjacent medium, while the guided mode must still remain sensitive to the variations of the dielectric constant of the superstrate. As shown in Fig. 3, the transverse and longitudinal modes were indeed not affected by placing ethanol vapour (refractive index

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1.36) in contact with the alumina-embedded nanorods if the nanorods are completely covered in matrix and slightly influenced by it when a nanoshell around them is created. However, the vapour injection led to a complete suppression of the guided mode as a result of the system being driven beyond the resonant excitation conditions.

To demonstrate the potential of this metamaterial for sensing purposes, we connected the prism-based coupling block with a flow cell to allow an access of the liquid to the nanorod assembly. Aqueous solutions of glycerine of different concentrations were then pumped through the cell to provide minute changes of the bulk refractive index of the nanorod surroundings. Figure 4a shows that the metamaterial works similarly to a conventional SPP-based sensor, showing a redshift of the resonance in response to an increase in the refractive index. Furthermore, a change of the refractive index by 10^{-4} RIU causes a shift of the resonance by 3.2 nm even without any optimization of the structure (see Fig. 4a, inset). The corresponding minimum estimation of sensitivity of 32,000 nm per RIU exceeds the sensitivity of localized plasmon-based schemes by two orders of magnitude[3,4]. The sensitivity of a guided wave offered by the nanorod metamaterial also exceeds the relevant parameter for commercial SPP-based sensors using spectral interrogation. Indeed, a similar variation of the refractive index by 10^{-4} causes a shift of the resonant wavelength by less than 0.2 nm (ref. 23), which is more than ten times lower than in the case of the nanorod metamaterial. An important comparative parameter of sensing devices is the figure of merit $\mathrm{FOM} = (\Delta\lambda/\Delta n)(1/\Delta\omega)$ (where $\Delta\omega$ is the full-width of the resonant dip at half-maximum and $\Delta\lambda$ is the resonance shift for Δn refractive-index change), which takes into account the sharpness of the resonance and thus examines the ability to sensitively measure tiny wavelength changes[3]. This parameter reaches 8 for LSP-based sensors[3,24] and 23 for the sensors based on SPP modes[3]. In the case of our metamaterial-based sensor, the minimum estimation of this parameter exceeds all reported figures of merit and reaches a value of 330 and higher. We reason that the origin of such a high sensitivity gain is twofold. In nanorod arrays, the sensed substance is incorporated between the initially bare rods, and therefore the waveguide mode provides a better overlap between the sensing field and the sensed substance than SPR sensors. Furthermore, the effective dielectric constant $\varepsilon_{\mathrm{eff}}$ of the metamaterial strongly depends on the dielectric constant of the tested medium ε_{d} as a result of the modification of the plasmon–plasmon interaction in the nanorod array[18–22], thus leading to modification of resonant conditions of the guided-mode excitation caused by the sensed analyte.

The second key advantage of the metamaterial-based sensor consists of the essentially discontinuous porous nanotexture of the nanorod matrix. This enables the implementation of new sensing geometries and strategies, not feasible with conventional film-based SPR. Indeed, by functionalizing the nanorods and immobilizing a receptor on their surface, one can follow the binding of a selective analyte with the receptor inside the nanorod matrix. The considerably increased surface area given by the nanoporous texture of the metamaterial significantly increases the amount of biomaterial that can be incorporated into the matrix within the available probe depth, maximizing the 'biological' sensitivity of the system. Furthermore, the distance between the nanorods can be selected to match the size of biological species of interest, giving access to a further size selectivity option that is important for many tasks in immunoassays and virus and protein detection[7,17].

To examine the feasibility of biosensing using the metamaterial-based transducer, we carried out tests using the streptavidin–biotin affinity model. Biotin is an analyte (vitamin) with a very small

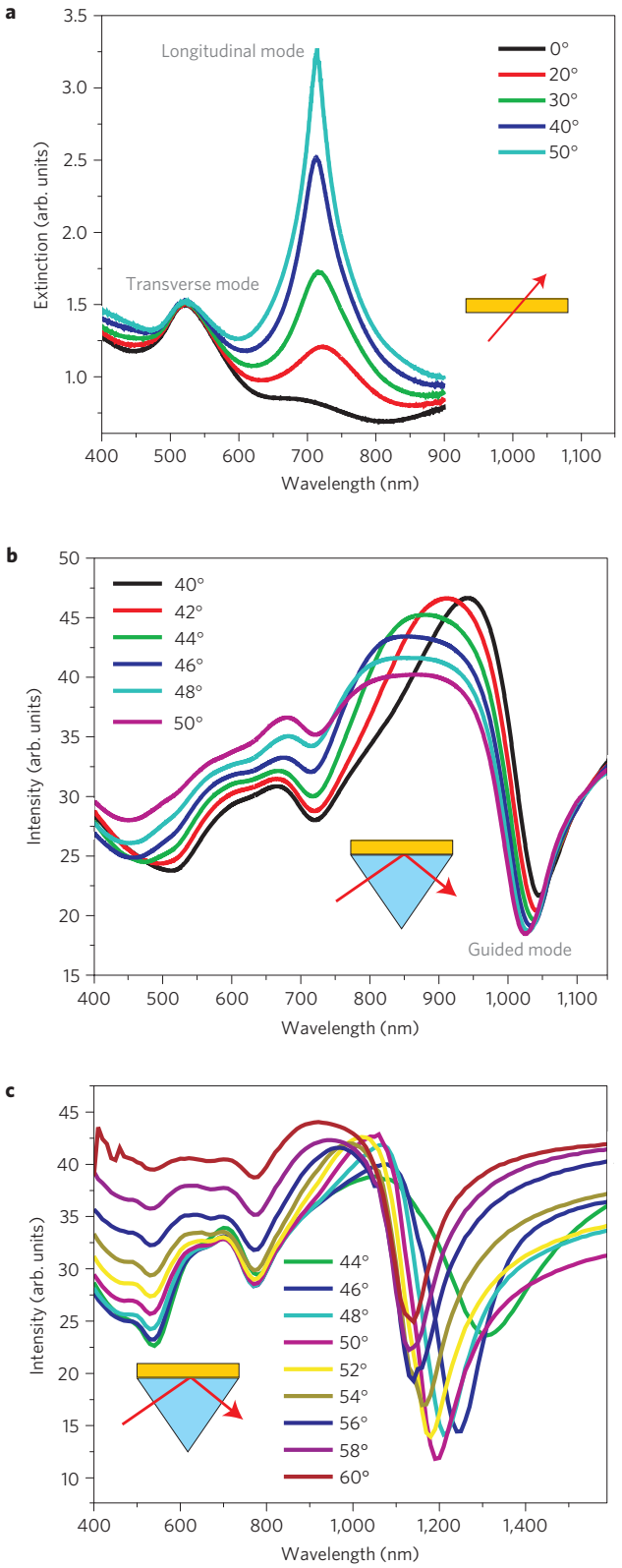

Figure 2 | Optical resonances of the plasmonic nanorod metamaterial in different surroundings. a, Extinction spectra of the nanorod array in air obtained in the direct-transmission geometry for different angles of illumination. **b**,**c**, Reflection spectra of the nanorod array in an air ($n = 1$) (**b**) and water environment ($n = 1.33$) (**c**), obtained in the ATR geometry for different angles of incidence. The nanorod array parameters are 380 nm length, 25 nm diameter and 60 nm spacing between the rods (centre-to-centre).

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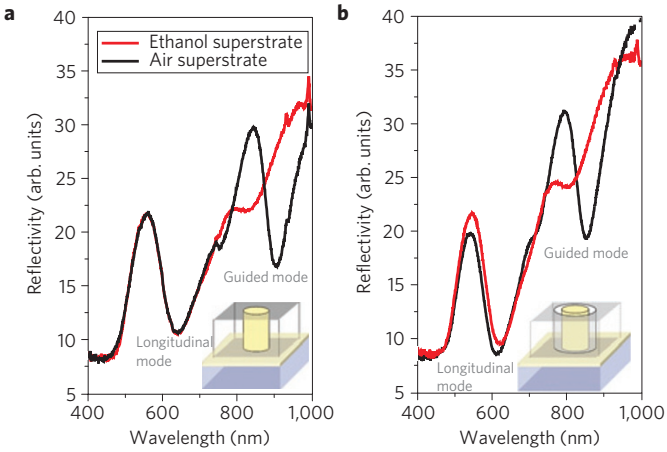

Figure 3 | Investigation of the origin of the sensing mode. a,**b**, ATR spectra of the nanorod arrays with different superstrates: air ($n = 1$) and ethanol ($n = 1.36$) for the nanorods in an alumina matrix (**a**) and with a 2 nm air-shell around the nanorods (**b**).

molecular weight (244 Da). The response of the system to this analyte can show the potential of the metamaterial for the detection of small compounds (drugs, vitamins, hormones and so on), for which conventional SPR devices are not sensitive enough. As a result of its small size, the detection of biotin by conventional SPR requires relatively high concentrations of 100 μM and higher[25]. To be able to provide a correct comparison with other plasmonic sensors, a standard analytical chemistry protocol needs to be used. In our tests, we strictly followed a biosensing protocol for the commercially available Thermo SPR system, described in ref. 25. After several functionalization steps, including the immobilization of 8-amino-1-octanethiol to make possible the attachment of streptavidin molecules and block unspecific sorption, the nanorods were immobilized by a streptavidin–melamide complex as a receptor. Owing to the fact that the functionalization protocol lasts 36 h and is well known, we did not record the transducer response to the preliminary functionalization steps. Before the injection of biological substances, the metamaterial was in contact with PBS buffer solution, and then biotin of different concentrations was injected into the flow cell. As shown in Fig. 4b, the biotin–streptavidin binding led to an immediate increase of the signal until its saturation, corresponding to a dynamic equilibrium of the binding process for the injected concentration of biotin. To remove unbound and weakly attached biotin, PBS buffer was injected, resulting in a certain decrease of the signal and its stabilization. Taking into account the level of noise in the system after averaging over the 0.05 nm range, the detection limit of the metamaterial-based sensor to biotin can be estimated to be below 300 nM, which is more than two orders of magnitude better than conventional SPR sensors using continuous films. (Note that the detection limit of large analytes such as streptavidin of 60,000 Da molecular weight can be in the picomolar range[26] owing to the mass and size scaling factor.) Such gain in sensitivity is due to a combination of the higher sensitivity of the metamaterial to bulk refractive-index change and a mass accumulation effect as a result of the large surface area of our nanoporous matrix. It is important that this scheme allowed us to follow the course of interactions in real time and hence resolve kinetic constants. As shown in Fig. 4, the biotin–streptavidin reaction had its own kinetics constant, describing the intensity of interaction. In fact, a routine ultra-sensitive determination of these constants is one of the main advantages of label-free plasmonics technology over fluorescent labelling methods.

The concept of a designer-made transducer with an enhanced sensing functionality, a porous bioimmobilization template

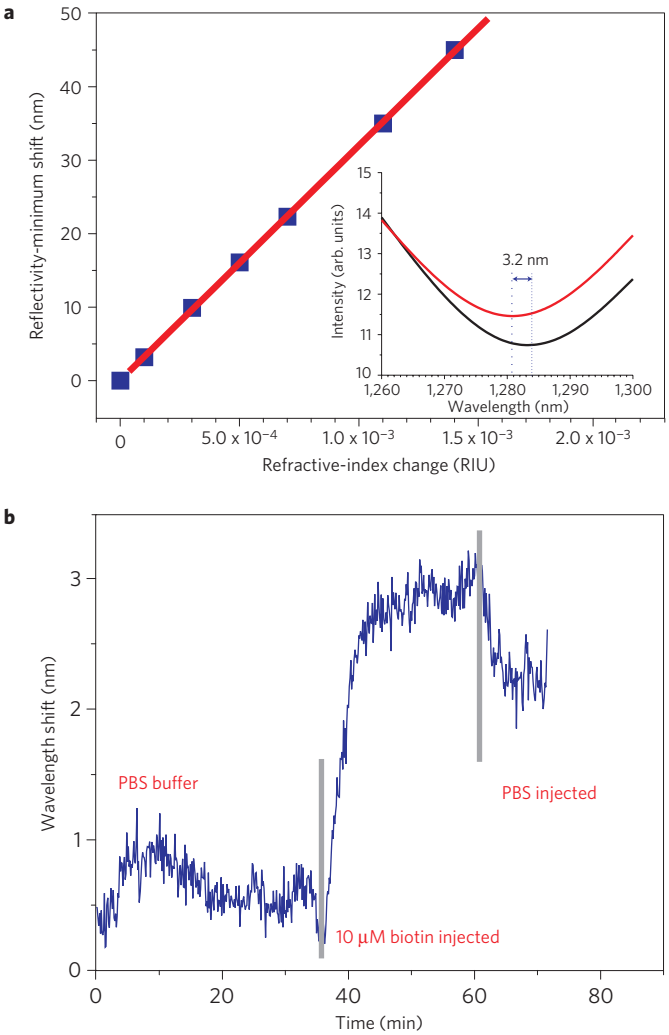

Figure 4 | The nanorod-based sensor calibration and observation of the test binding reaction. a, Calibration curve for the metamaterial-based sensor under the step-like changes of the refractive index of the environment using different glycerine–water solutions. The measurements were carried out at a wavelength of 1,230 nm. The size of the squares represents error bars. Inset: Reflectivity spectrum modifications with the changes of the refractive index by 10^{-4} RIU. **b**, Response of the metamaterial-based biosensor to the reaction of biotin–streptavidin binding.

with controlled size of pore openings, and wavelength-tunable properties offers numerous opportunities in the development of new biological and chemical sensors and sensing strategies. The possibility to design the working wavelength and sensitivity by changing the nanostructure parameters makes the proposed transducer a universal tool for applications in the sensing of biological and chemical molecules of different properties, for example, size and/or absorption, and yet compatible with the existing SPR equipment. The nanoporous texture of the metamaterial further promises an improved sensitivity owing to the possibility of increasing the amount of tested biomaterial, size selectivity and further field-enhancement-based applications for molecular identification, such as surface-enhanced fluorescence and/or surface-enhanced Raman scattering, inside the nanorod matrix. Finally, as the metamaterial fabrication procedure is simple and compatible with microchips and microfluidic technologies, the metamaterial-based biosensor can be integrated into complex systems such as micro total-analysis units.

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Received 18 December 2008; accepted 14 September 2009; published online 11 October 2009

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Acknowledgements

This work was supported in part by The Royal Society and the EPSRC (UK). V.A.P. acknowledges support from ONR (Grant No. N00014-07-1-0457) and NSF (Grant No. ECCS-0724763). The authors are grateful to T. W. Odom for insightful discussions of sensing parameters.

Author contributions

All authors contributed extensively to the work presented in this letter.

Additional information

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